

Assignment No 7

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Aim: (Functions)

Write a Python function to -

- find the Max of three numbers
- multiply all the numbers in a list
- create a list of all even numbers between 19 and 88
- display the current date and time and get the Python version

Theory:

Python Functions is a block of statements that return the specific task.

Creating a Python Function :

We can create a Python function using the def keyword.

def fun():

 print("Welcome to Python Function")

Calling a Python Function

After creating a function we can call it by using the name of the function followed by parenthesis containing parameters of that particular function.

def fun():

 print("Welcome to Python Function")

#driver code

fun() #function call

OUTPUT: Welcome to Python Function

Using arguments:

def add(num1: int, num2: int) -> int:

 """Add two numbers"""

```
num3 = num1 + num2
```

```
return num3
```

```
# Driver code
```

```
num1, num2 = 5, 15
```

```
ans = add(num1, num2)
```

```
print(f"The addition of {num1} and {num2} results {ans}.")
```

Output: The addition of 5 and 15 results 20.

For date time and current version of python :

Code:

```
import datetime
```

```
current_time=datetime.datetime.now()
```

```
print ("date and time are:", current_time)
```

```
import sys
```

```
print("Python version: ", sys.version)
```

Output: date and time are: 2022-11-16 15:30:12.554061

Python version: 3.10.8 (tags/v3.10.8:aaaf517, Oct 11 2022, 16:50:30) [MSC v.1933 64 bit (AMD64)]

Program:

```
#1. Find the max of three numbers

def max_of_3 ():

    a=int (input("Enter the value of a"))

    b=int (input("Enter the value of b"))

    c=int (input("Enter the value of c"))

    if a>b and a>c:

        print ("a is greater")
```

```
elif b>a and b>c:

    print ("b is greater")

elif c>a and c>b:

    print ("c is greater")

else:

    print("All three numbers all equal")

def list_mul ():

    n=int (input("Enter the number of element you want in the list ::
"))

    i=0

    l1=[]

    for i in range (0,n):

        element=int (input ("Enter the element:"))

        l1.append(element)

    print ("List entered by you is: ")

    print (l1)

    mult=1

    for i in range (0,n):

        mult=mult*l1[i]

    print ("Multiplication of the elements of the List is ",mult)

def even_num ():

    l2=[]

    for i in range (19,89):

        if i%2==0:

            l2.append(i)

    print ("Even numbers between 19 and 88 are :")

    print(l2)

def date_time():

    import datetime
```

```

current_time=datetime.datetime.now()

print ("date and time are:", current_time)

import sys

print("Python version: ", sys.version)

def menu():

    print ("1.Max of three numbers \n2. Multiply numbers within a list
\n3.Make a list of all even numbers between 18 and 88 \n4.Print current
time date and version of python \n0.Exit")

menu ()

option=int(input ("Enter the option number :"))

while (option!=0):

    if option==1:

        max_of_3()

    elif option==2:

        list_mul()

    elif option==3:

        even_num()

    elif option==4:

        date_time ()

    menu()

    option=int (input("Enter the option number :"))

print ("Exiting.....")

```

Output:

```

Rudradev Arya@LAPTOP-24G9TENL MINGW64 ~
$ "C:/Users/Rudradev
Arya/AppData/Local/Programs/Python/Python310/python.exe"

```

```
"c:/Users/Rudradev Arya/Desktop/Rudradev/MIT ADT/SY Sem  
1/PPL/MITADTSY/ui_in_python.py"
```

- 1.Max of three numbers
- 2. Multiply numbers within a list
- 3.Make a list of all even numbers between 18 and 88
- 4.Print current time date and version of python
- 0.Exit

Enter the option number :1

Enter the value of a2

Enter the value of b3

Enter the value of c4

c is greater

- 1.Max of three numbers
- 2. Multiply numbers within a list
- 3.Make a list of all even numbers between 18 and 88
- 4.Print current time date and version of python
- 0.Exit

Enter the option number :2

Enter the number of element you want in the list :: 3

Enter the element:2

Enter the element:3

Enter the element:4

List entered by you is:

```
[2, 3, 4]
```

Multiplication of the elements of the List is 24

- 1.Max of three numbers
- 2. Multiply numbers within a list
- 3.Make a list of all even numbers between 18 and 88
- 4.Print current time date and version of python

```
0.Exit
Enter the option number :3
Even numbers between 19 and 88 are :
[20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52,
54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88]
1.Max of three numbers
2. Multiply numbers within a list
3.Make a list of all even numbers between 18 and 88
4.Print current time date and version of python
0.Exit
Enter the option number :4
date and time are: 2022-11-17 22:21:45.014999
Python version: 3.10.7 (tags/v3.10.7:6cc6b13, Sep 5 2022, 14:08:36)
[MSC v.1933 64 bit (AMD64)]
1.Max of three numbers
2. Multiply numbers within a list
3.Make a list of all even numbers between 18 and 88
4.Print current time date and version of python
0.Exit
Enter the option number :0
Exiting.....
```

Conclusion: We have successfully learnt how to create define and use functions in python also to perform various operations on lists and its contents.